Winning Space Race with Data Science

Predicting SpaceX Falcon 9 first-stage landing success

IBM Data Science Professional Certificate — Applied Data Science Capstone

GitHub: https://github.com/TrungKiencding/IBM-Data-Science-Capstone-SpaceX

September 2026

Outline

- Executive Summary
- Introduction
- Methodology
- Results
  - Exploratory data analysis (visualization & SQL)
  - Interactive analytics (Folium map, Plotly Dash)
  - Predictive analysis (classification)
- Conclusion
- Appendix

Executive Summary

- Methodologies
  - Data collection via the SpaceX REST API and web scraping of Wikipedia's Falcon 9 launch tables
  - Data wrangling: missing-value handling, one-hot encoding, creation of the binary landing label 'Class'
  - EDA with pandas/Matplotlib/Seaborn and SQL; interactive analytics with Folium and Plotly Dash
  - Predictive modelling: Logistic Regression, SVM, Decision Tree and KNN tuned with 10-fold GridSearchCV
- Results
  - Overall landing success rate is 67%; it rose from 0% (2010–2012) to over 80% after 2017
  - KSC LC-39A has the most successful launches; ES-L1, GEO, HEO and SSO orbits reached 100% success
  - All four models reach ~83% test accuracy; Decision Tree was selected (CV accuracy 87.7%) — main error mode is false positives

Introduction

- Project background and context
  - SpaceX advertises Falcon 9 launches at ~62*million*, *whilecompetitorschargeupwardof*165 million.
  - Most of the saving comes from re-using the first stage. If we can predict whether the first stage will land, we can estimate the cost of a launch — valuable for a competitor ('SpaceY') bidding against SpaceX.
- Problems we want to answer
  - Which factors (payload, orbit, launch site, booster version, flight number…) influence landing success?
  - How do these factors interact, and how has success evolved over time?
  - What operating conditions are needed for a successful landing program?
  - Can a machine-learning model reliably predict landing success for a new launch?

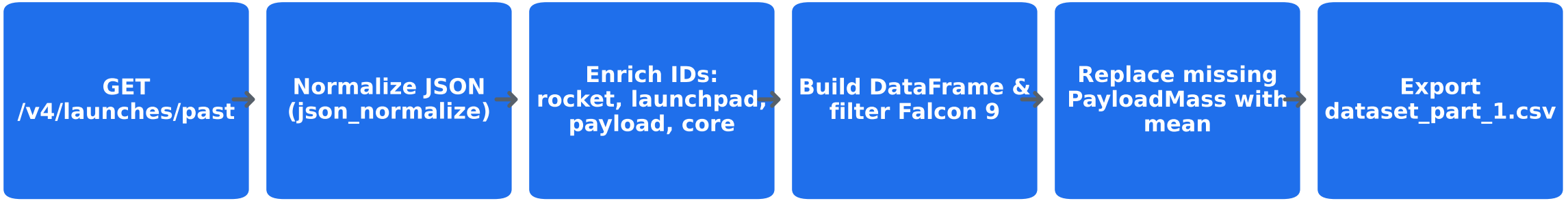
Section 1

Methodology

Methodology

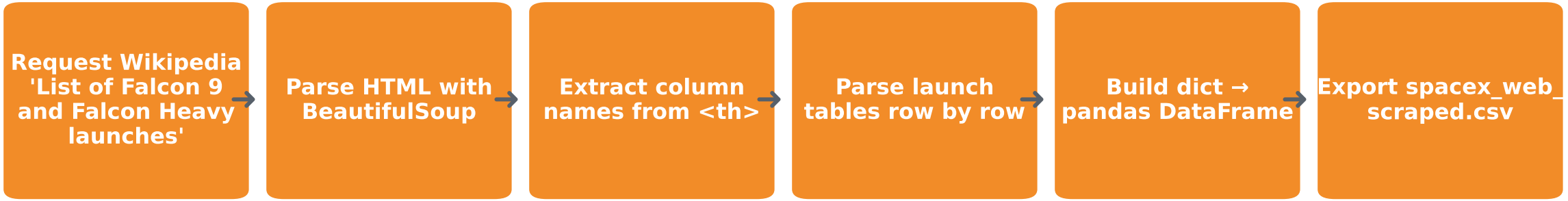
- Data collection: SpaceX REST API (api.spacexdata.com/v4) + web scraping of Wikipedia with BeautifulSoup
- Data wrangling: filtered to Falcon 9, replaced missing PayloadMass with the mean, derived 'Class' label from 'Outcome' (True ASDS / RTLS / Ocean → 1, others → 0)
- Exploratory data analysis (EDA) using visualization (scatter, bar, line plots) and SQL (SQLite / Db2)
- Interactive visual analytics using Folium (launch-site maps, marker clusters, distance lines) and Plotly Dash
- Predictive analysis using classification models
  - Standardize features → train/test split (80/20) → GridSearchCV (cv=10) → compare accuracy & confusion matrices

Data Collection – SpaceX API



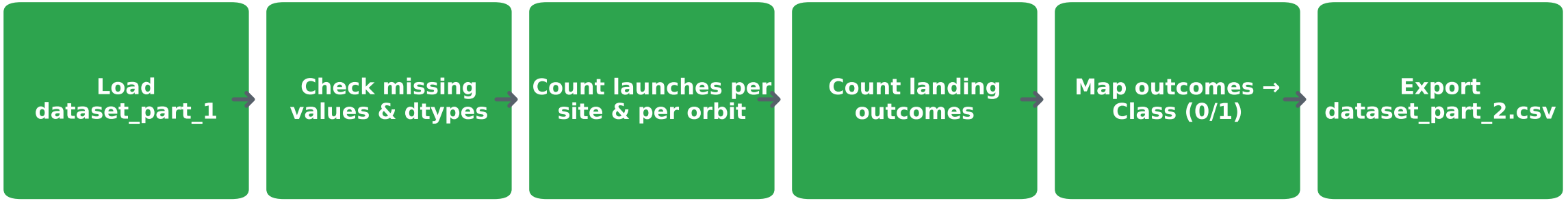
- Helper functions getBoosterVersion(), getLaunchSite(), getPayloadData(), getCoreData() call extra endpoints to resolve IDs into readable values.
- Result: 90 Falcon 9 launches with 17 columns (FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, Latitude).
- Notebook: https://github.com/TrungKiencding/IBM-Data-Science-Capstone-SpaceX/blob/main/1_jupyter-labs-spacex-data-collection-api.ipynb

Data Collection – Web Scraping



- Captured Flight No., Date, Time, Version Booster, Launch Site, Payload, Payload mass, Orbit, Customer, Launch outcome and Booster landing for 121 launches.
- Notebook: https://github.com/TrungKiencding/IBM-Data-Science-Capstone-SpaceX/blob/main/2_jupyter-labs-webscraping.ipynb

Data Wrangling



- Landing outcomes: True ASDS, True RTLS, True Ocean = success (1); False ASDS, False RTLS, False Ocean, None ASDS, None None = failure (0).
- Resulting average success rate: 66.7%.
- Notebook: https://github.com/TrungKiencding/IBM-Data-Science-Capstone-SpaceX/blob/main/3_labs-jupyter-spacex-Data-wrangling.ipynb

EDA with Data Visualization

- Scatter plots — relationship between two variables coloured by landing outcome:
  - Flight Number vs Launch Site, Payload vs Launch Site, Flight Number vs Orbit, Payload vs Orbit
- Bar chart — success rate of each orbit type (compare categorical groups)
- Line chart — yearly average success rate (show trend over time)
- Feature engineering: one-hot encoding of Orbit, LaunchSite, LandingPad, Serial → 83 numeric features (dataset_part_3.csv)
- Notebook: https://github.com/TrungKiencding/IBM-Data-Science-Capstone-SpaceX/blob/main/5_edadataviz.ipynb

EDA with SQL

- Loaded Spacex.csv into a SQL database (SQLite, table SPACEXTBL) and queried with %sql magic:
  - Unique launch sites; 5 records whose site begins with 'CCA'
  - Total payload of NASA (CRS) missions; average payload of booster F9 v1.1
  - Date of first successful ground-pad landing
  - Boosters with drone-ship success and payload 4000–6000 kg
  - Count of successful / failed mission outcomes
  - Boosters that carried the maximum payload (sub-query)
  - 2015 drone-ship failures by month; ranking of landing outcomes 2010-06-04 → 2017-03-20
- Notebook: https://github.com/TrungKiencding/IBM-Data-Science-Capstone-SpaceX/blob/main/4_jupyter-labs-eda-sql-coursera_sqllite.ipynb

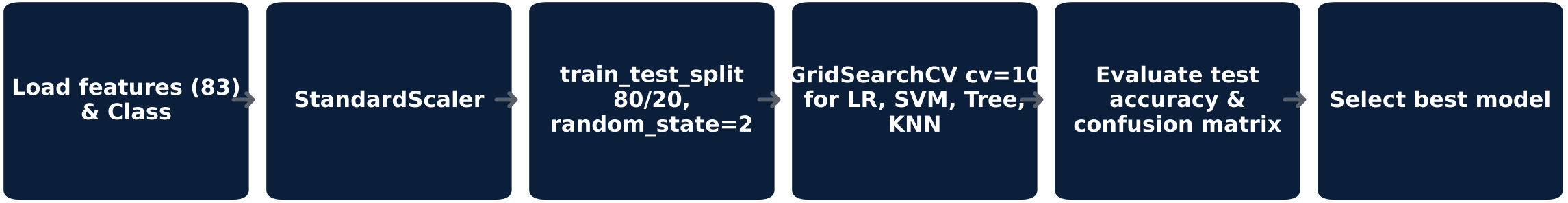
Build an Interactive Map with Folium

- Circles + text markers for each launch site at its latitude/longitude (NASA JSC as reference point)
- MarkerCluster with green (success) / red (failure) markers for every launch record at each site
- MousePosition plugin to read coordinates, and PolyLines with distance labels (Haversine) from a launch site to the nearest coastline, railway, highway and city
- Why: to see whether launch sites are near the equator/coast and how proximity relates to safety & logistics
- Notebook: https://github.com/TrungKiencding/IBM-Data-Science-Capstone-SpaceX/blob/main/6_lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- Dropdown to select All Sites or an individual launch site
- Pie chart: total successful launches by site (All) or success vs failure for a chosen site
- RangeSlider to filter payload mass (0 – 10 000 kg)
- Scatter chart: payload mass vs outcome, coloured by booster version category
- Why: lets stakeholders interactively explore which sites, payload ranges and boosters give the highest success
- Code: https://github.com/TrungKiencding/IBM-Data-Science-Capstone-SpaceX/blob/main/spacex_dash_app.py

Predictive Analysis (Classification)



- Hyper-parameters searched: LR (C), SVM (kernel, C, gamma), Decision Tree (criterion, splitter, depth, leaf/split sizes), KNN (n_neighbors, algorithm, p).
- Notebook: https://github.com/TrungKiencding/IBM-Data-Science-Capstone-SpaceX/blob/main/7_SpaceX_Machine_Learning_Prediction_Part_5.ipynb

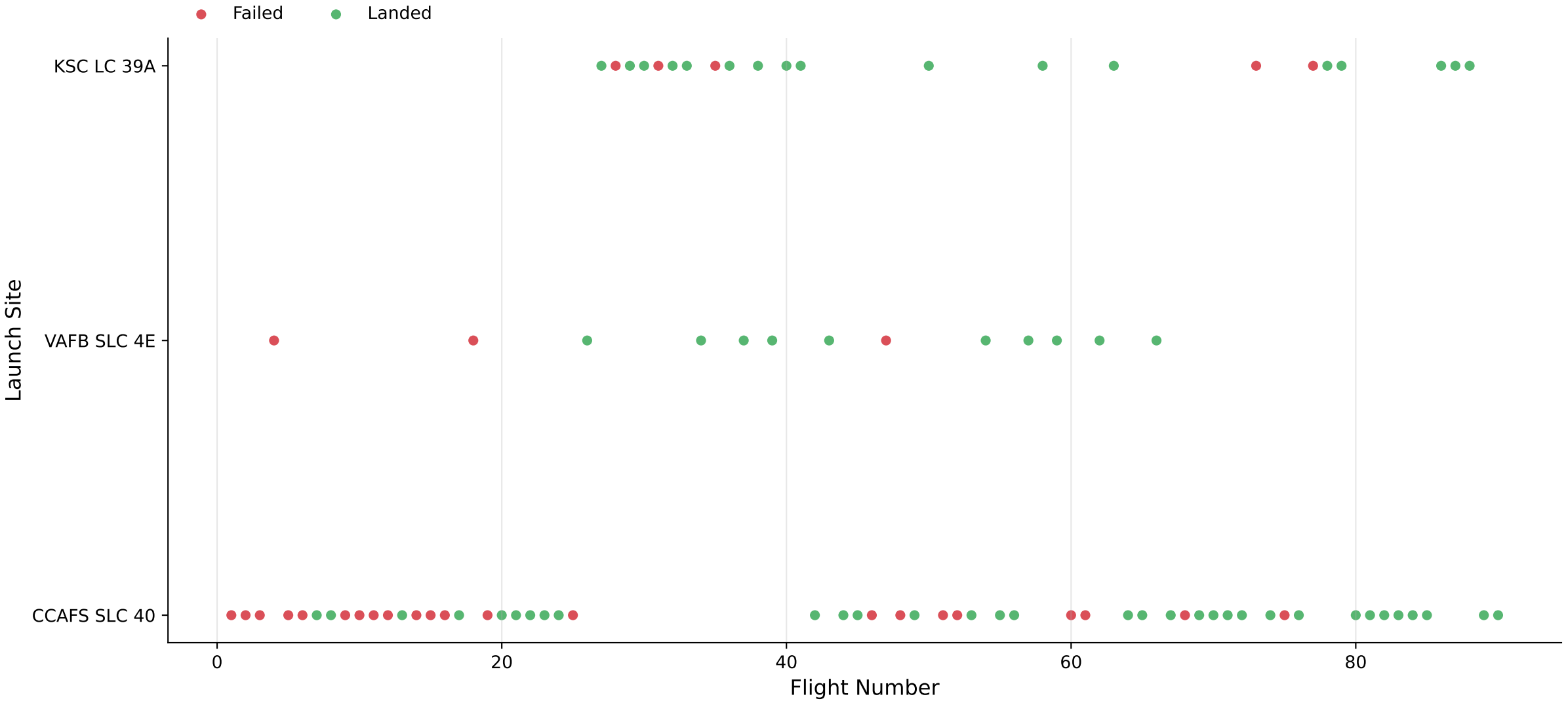
Results

- Exploratory data analysis: success rate improves strongly with flight number/experience; 67% overall
- Interactive analytics: all sites are close to the coast and far from cities; KSC LC-39A is the most successful site
- Predictive analysis: tuned classifiers reach 83.3% test accuracy; Decision Tree chosen as best model

Section 2

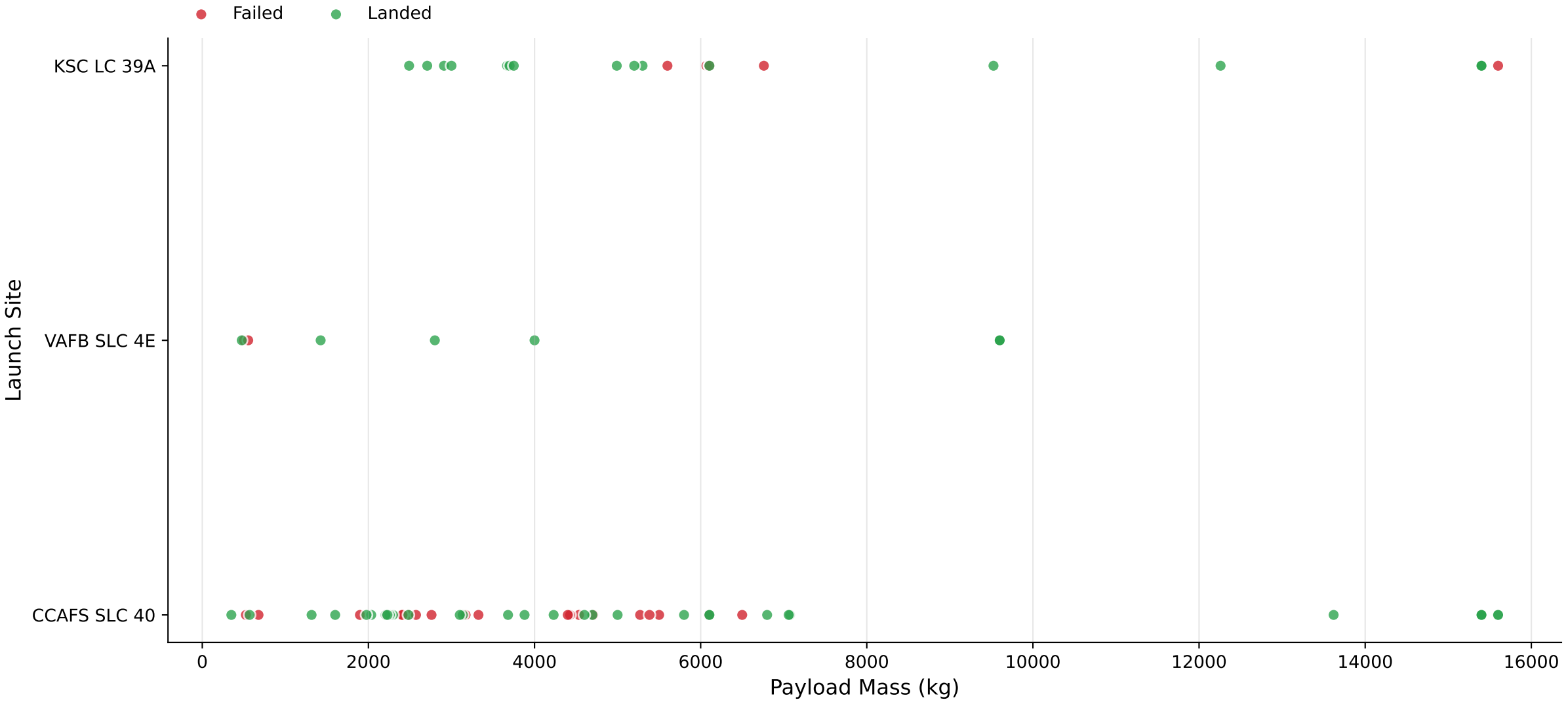
Insights drawn from EDA

Flight Number vs. Launch Site



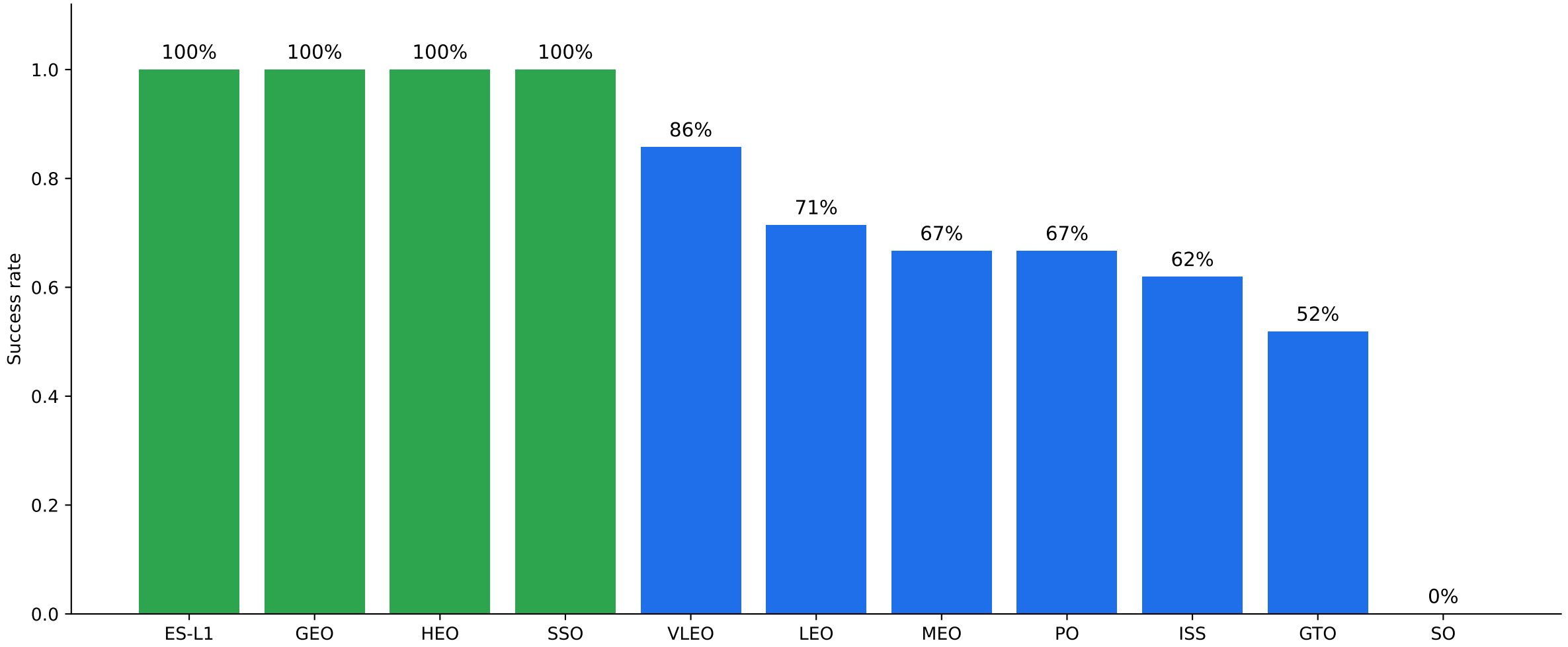
Insight: Early flights (mostly CCAFS SLC 40) failed; success grows with flight number at every site.

Payload vs. Launch Site



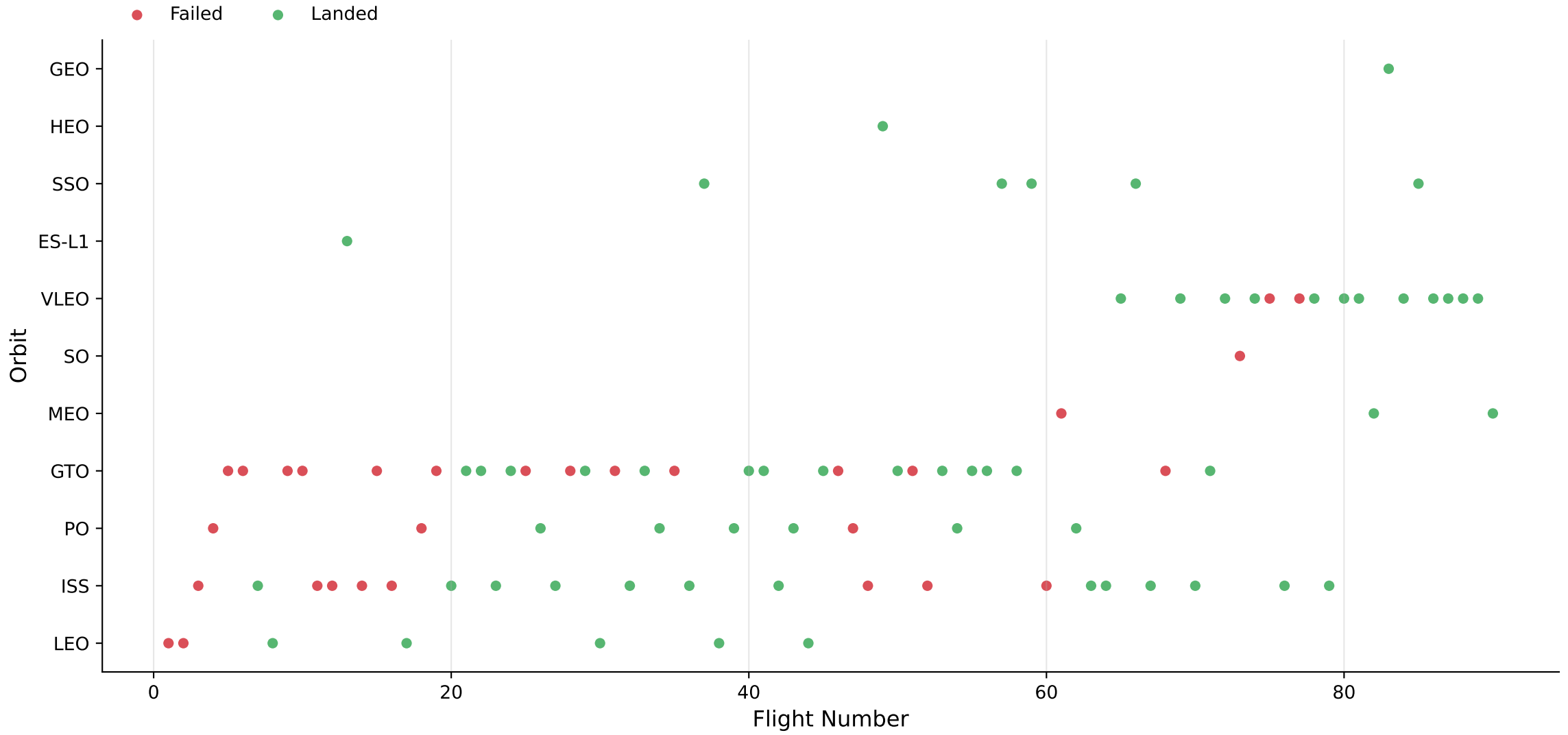
Insight: VAFB SLC 4E launched no payload > 10 000 kg; heavy payloads (>7 000 kg) mostly land successfully.

Success Rate vs. Orbit Type



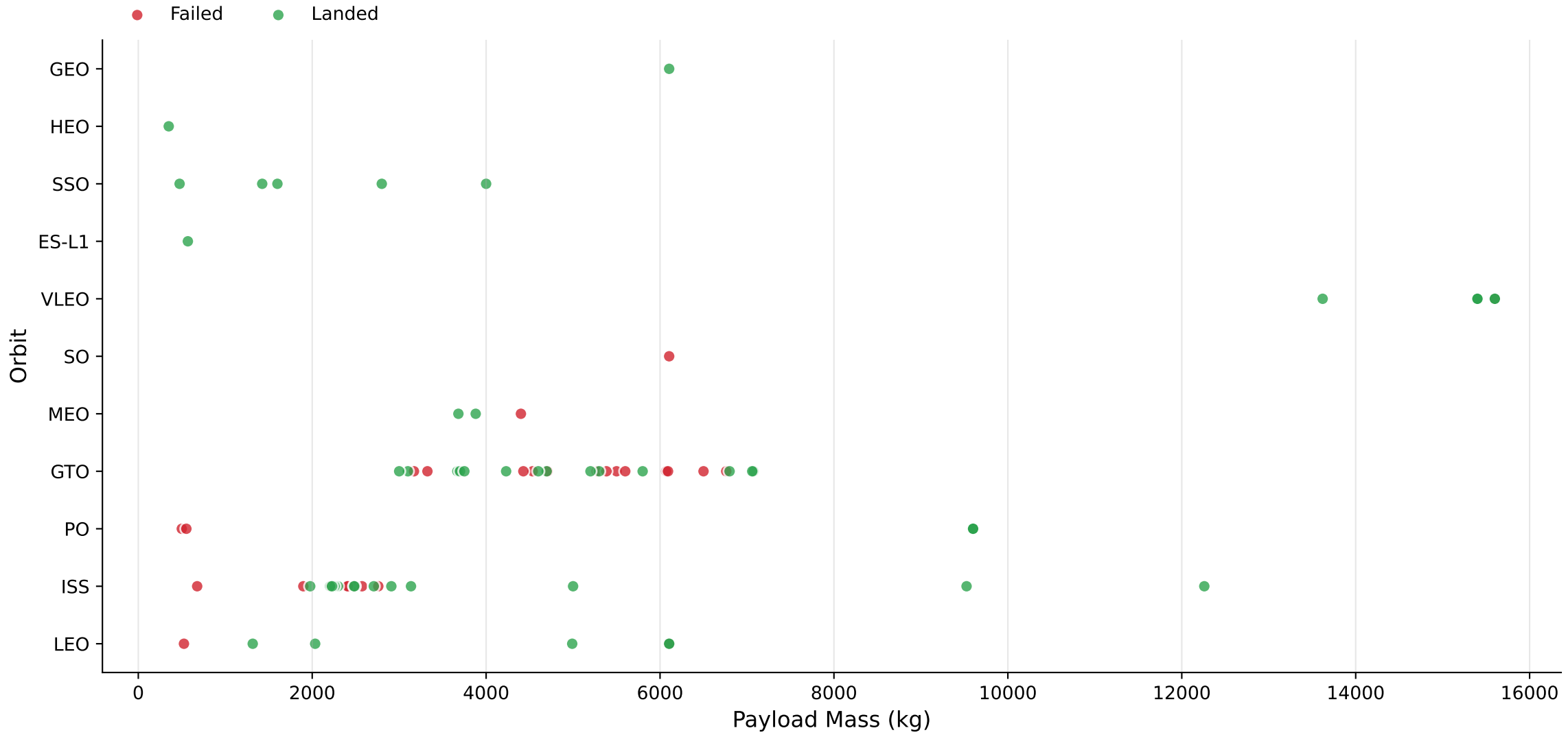
Insight: ES-L1, GEO, HEO and SSO reach 100% success; GTO (~52%) and SO (0%) are the hardest orbits.

Flight Number vs. Orbit Type



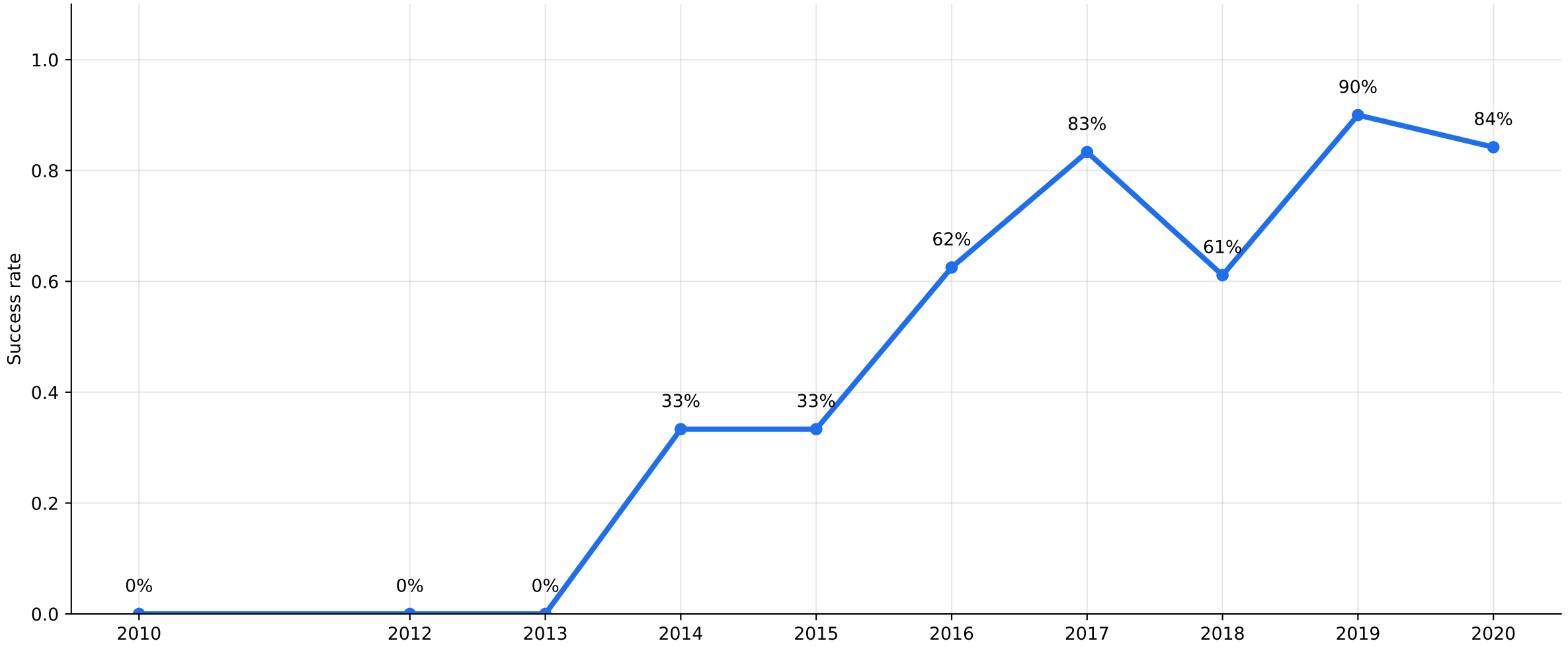
Insight: In LEO success follows flight number; in GTO there is no clear relationship with flight number.

Payload vs. Orbit Type



Insight: Heavy payloads have higher success for Polar, LEO and ISS; GTO mixes outcomes across payloads.

Launch Success Yearly Trend



Insight: Success rate kept rising from 2013 to 2020 (small dip in 2018) — SpaceX learns quickly.

All Launch Site Names


```
SELECT DISTINCT Launch_Site
FROM SPACEXTBL;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Explanation

- DISTINCT removes duplicates so each launch site appears once: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E.
- Rows returned: 4

Insight: There are four unique launch sites in the data.

Launch Site Names Begin with 'CCA'


```
SELECT * FROM SPACEXTBL
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

Date	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Landing_Outcome
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qu	0	LEO	SpaceX	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1	0	LEO (ISS)	NASA (COTS) NRO	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	No attempt

Insight: The first five CCA launches were all early LEO missions for SpaceX/NASA.

Total Payload Mass


```
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload
FROM SPACEXTBL
WHERE Customer = 'NASA (CRS)';
```

Total_Payload
45596

Explanation

- SUM aggregates PAYLOAD_MASS__KG_ over all rows whose customer is NASA (CRS) – the ISS resupply missions.
- Rows returned: 1

Insight: Total payload carried by boosters launched for NASA (CRS).

Average Payload Mass by F9 v1.1


```
SELECT AVG(PAYLOAD_MASS__KG_) AS Avg_Payload
FROM SPACEXTBL
WHERE Booster_Version = 'F9 v1.1';
```

Avg_Payload
2928.4

Explanation

- AVG computes the mean payload of the F9 v1.1 booster, an early version with modest lift capacity.
- Rows returned: 1

Insight: Average payload mass carried by booster version F9 v1.1.

First Successful Ground Landing Date


```
SELECT MIN(Date) AS First_Success
FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (ground pad)';
```

First_Success
2015-12-22

Explanation

- MIN(Date) on rows with 'Success (ground pad)' finds the first RTLS landing – a milestone for reusability.
- Rows returned: 1

Insight: The first successful ground-pad landing happened in December 2015.

Successful Drone Ship Landing with Payload 4000–6000


```
SELECT Booster_Version FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (drone ship)'
  AND PAYLOAD_MASS__KG_ > 4000
  AND PAYLOAD_MASS__KG_ < 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Explanation

- Combines the landing-outcome filter with a payload range using AND to list qualifying boosters.
- Rows returned: 4

Insight: Four F9 FT boosters landed on a drone ship with a 4000–6000 kg payload.

Total Number of Successful and Failure Mission Outcomes


```
SELECT Mission_Outcome, COUNT(*) AS Total
FROM SPACEXTBL
GROUP BY Mission_Outcome;
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear	1

Explanation

- GROUP BY Mission_Outcome with COUNT(*) shows mission (not landing) reliability is ~99%.
- Rows returned: 4

Insight: Almost every mission succeeded; only one in-flight failure.

Boosters Carried Maximum Payload


```
SELECT DISTINCT Booster_Version FROM SPACEXTBL
WHERE PAYLOAD_MASS__KG_ = (
  SELECT MAX(PAYLOAD_MASS__KG_)
  FROM SPACEXTBL);
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

Explanation

- A sub-query returns MAX(payload); the outer query lists every booster that carried it (15,600 kg).
- Rows returned: 12

Insight: All boosters carrying the max payload are Block 5 (F9 B5) versions.

2015 Launch Records


```
SELECT substr(Date,6,2) AS Month, Landing_Outcome,
       Booster_Version, Launch_Site
FROM SPACEXTBL
WHERE substr(Date,1,4) = '2015'
  AND Landing_Outcome = 'Failure (drone ship)';
```

Month	Landing_Outcom	Booster_Version	Launch_Site
01	Failure (drone sh	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone sh	F9 v1.1 B1015	CCAFS LC-40

Explanation

- substr() extracts month and year from the text date because SQLite has no month-name function.
- Rows returned: 2

Insight: Two drone-ship landing failures in 2015 (January and April), both from CCAFS LC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20


```
SELECT Landing_Outcome, COUNT(*) AS Cnt
FROM SPACEXTBL
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Cnt DESC;
```

Landing_Outcome	Cnt
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Explanation

- GROUP BY + ORDER BY COUNT DESC ranks landing outcomes in the early period of the program.
- Rows returned: 8

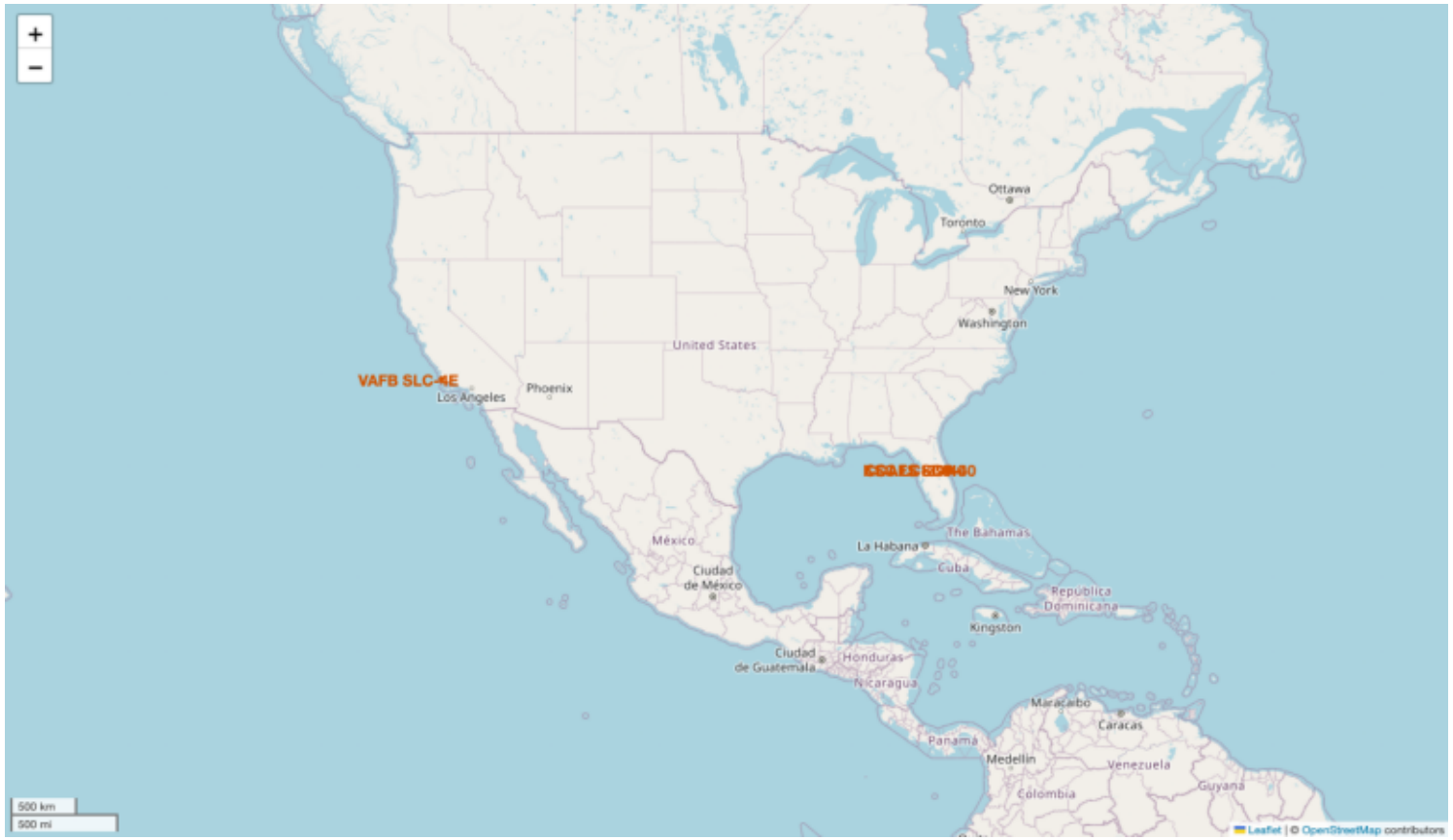
Insight: 'No attempt' dominates this early period, followed by drone-ship successes and failures.

Section 3

Launch Sites Proximities Analysis

All Launch Sites on a Global Map

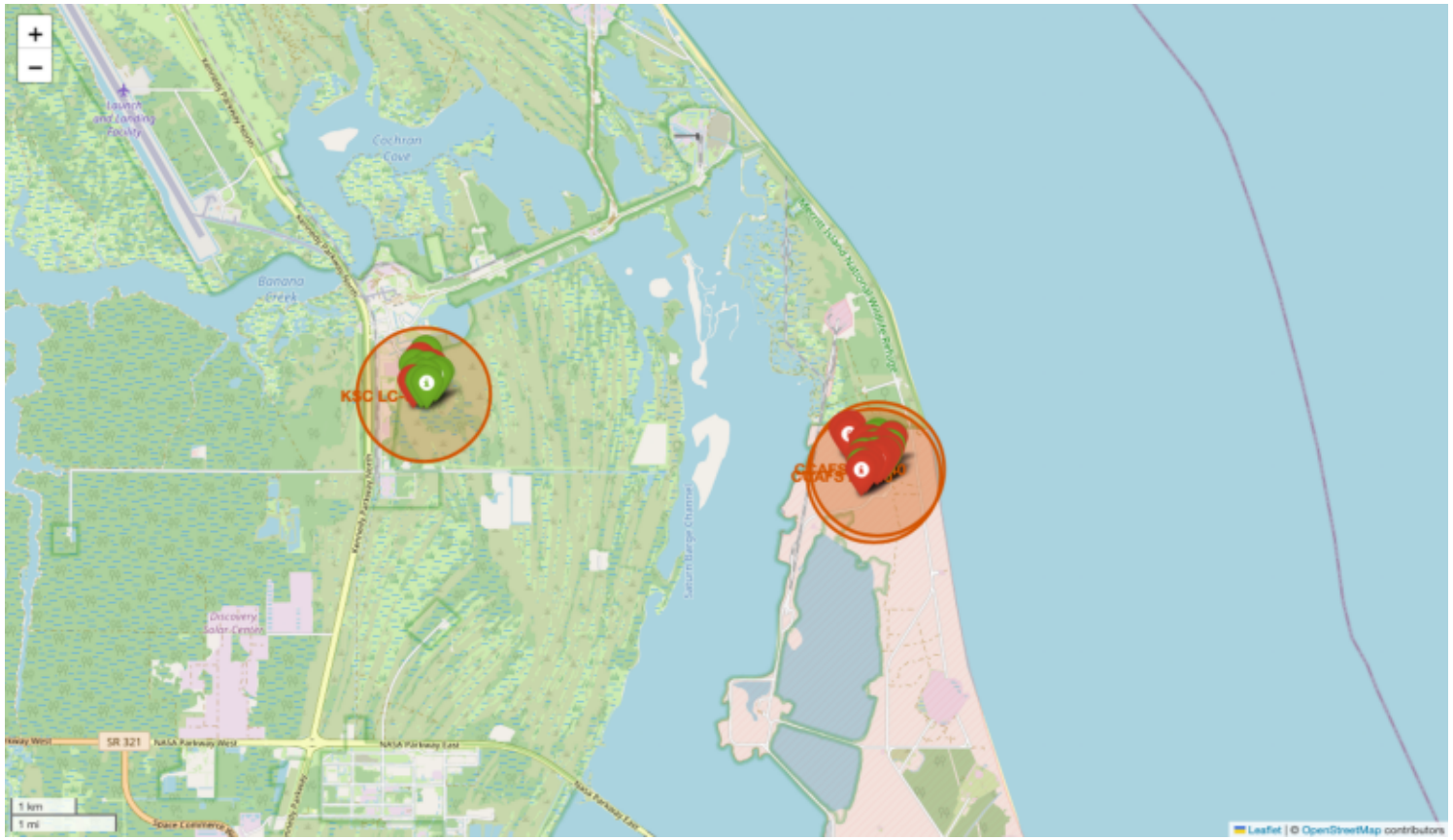
FOLIUM



- All 4 launch sites are in the USA: 3 in Florida (east coast) and VAFB SLC-4E in California (west coast).
- Sites are as far south as possible – closer to the equator, where Earth's rotation gives extra velocity and saves fuel.
- All sites are on the coast so failed rockets fall into the ocean, away from people.

Launch Outcomes per Site (Colour-labelled Markers)

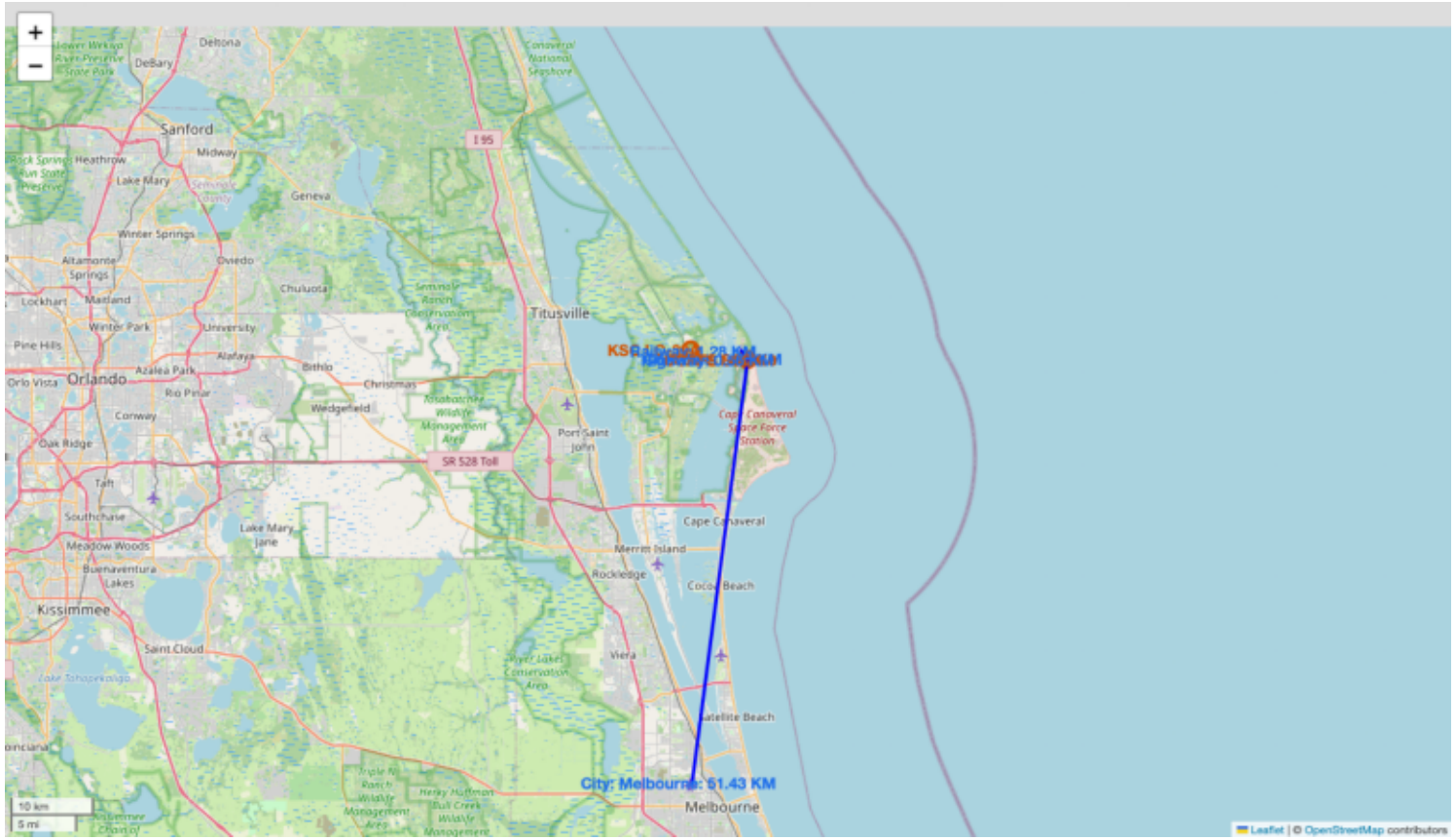
FOLIUM



- Each launch is a marker in a MarkerCluster: green = successful landing, red = failure.
- KSC LC-39A (left) shows mostly green markers – highest success ratio (76.9%).
- CCAFS LC-40 (right) has the most launches but many red markers, mainly early flights.

Distances from CCAFS SLC-40 to Proximities

FOLIUM

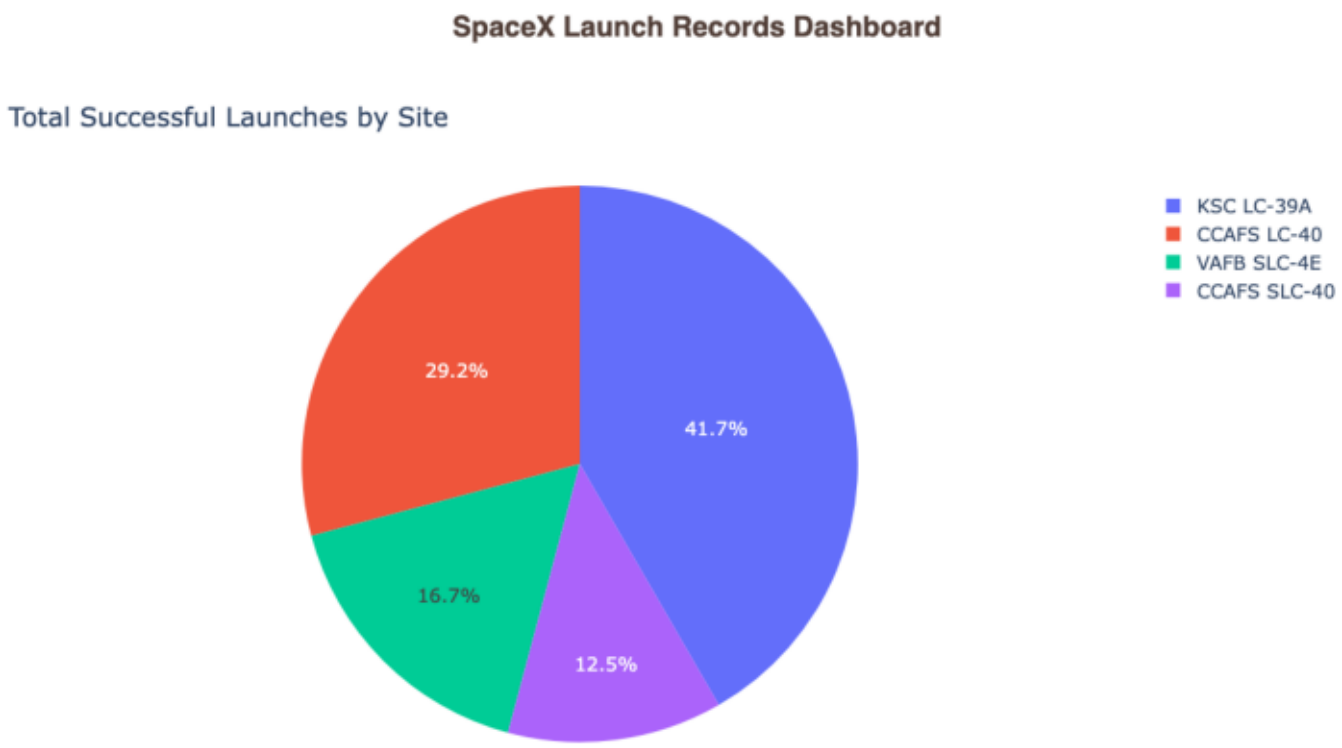


- Distances computed with the Haversine formula and drawn as PolyLines:
  - Coastline 0.86 km, Highway 0.56 km, Railway 1.28 km → easy transport of heavy parts and safe ocean trajectory.
  - Nearest city (Melbourne) 51.4 km → far from populated areas for safety.

Section 4

Build a Dashboard with Plotly Dash

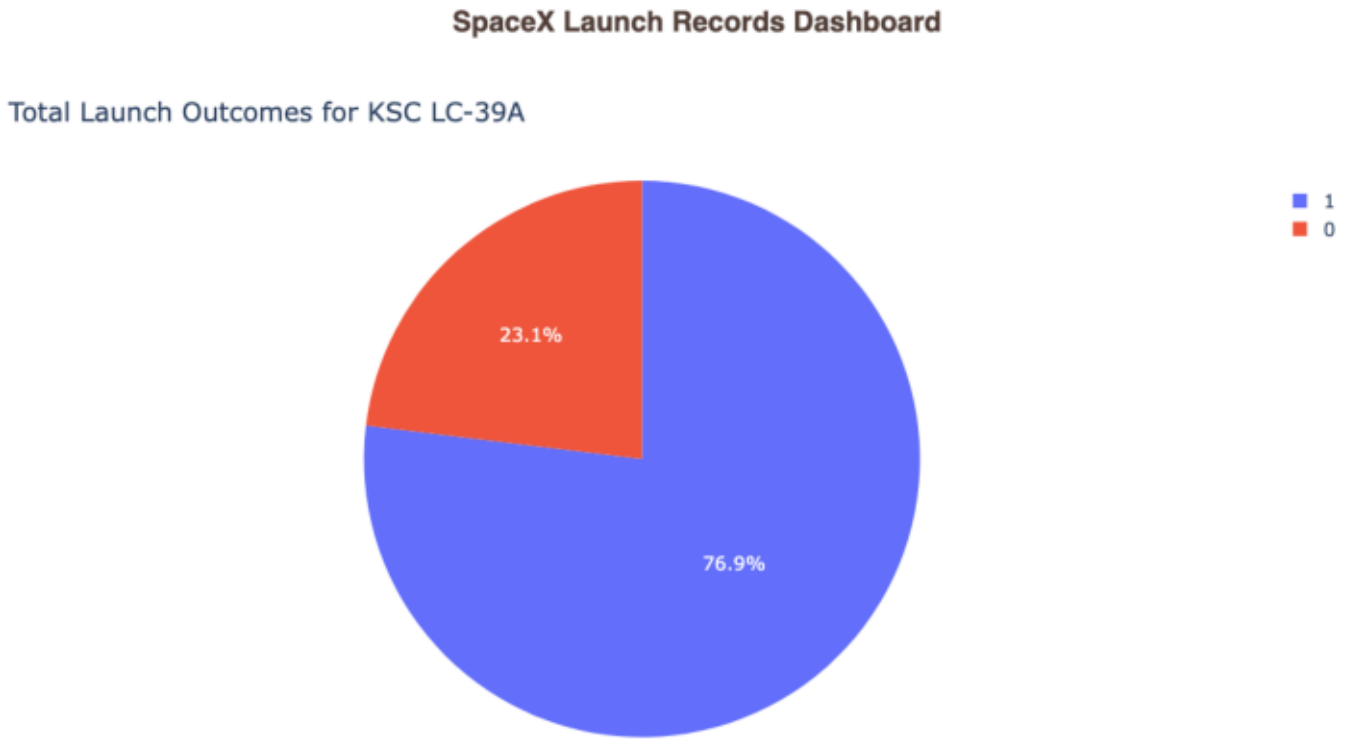
Launch Success Count for All Sites



- Dropdown = 'All Sites': pie of successful launches by site.
- KSC LC-39A has the largest share of successes (41.7%), followed by CCAFS LC-40 (29.2%).
- CCAFS SLC-40 has the smallest share (12.5%).

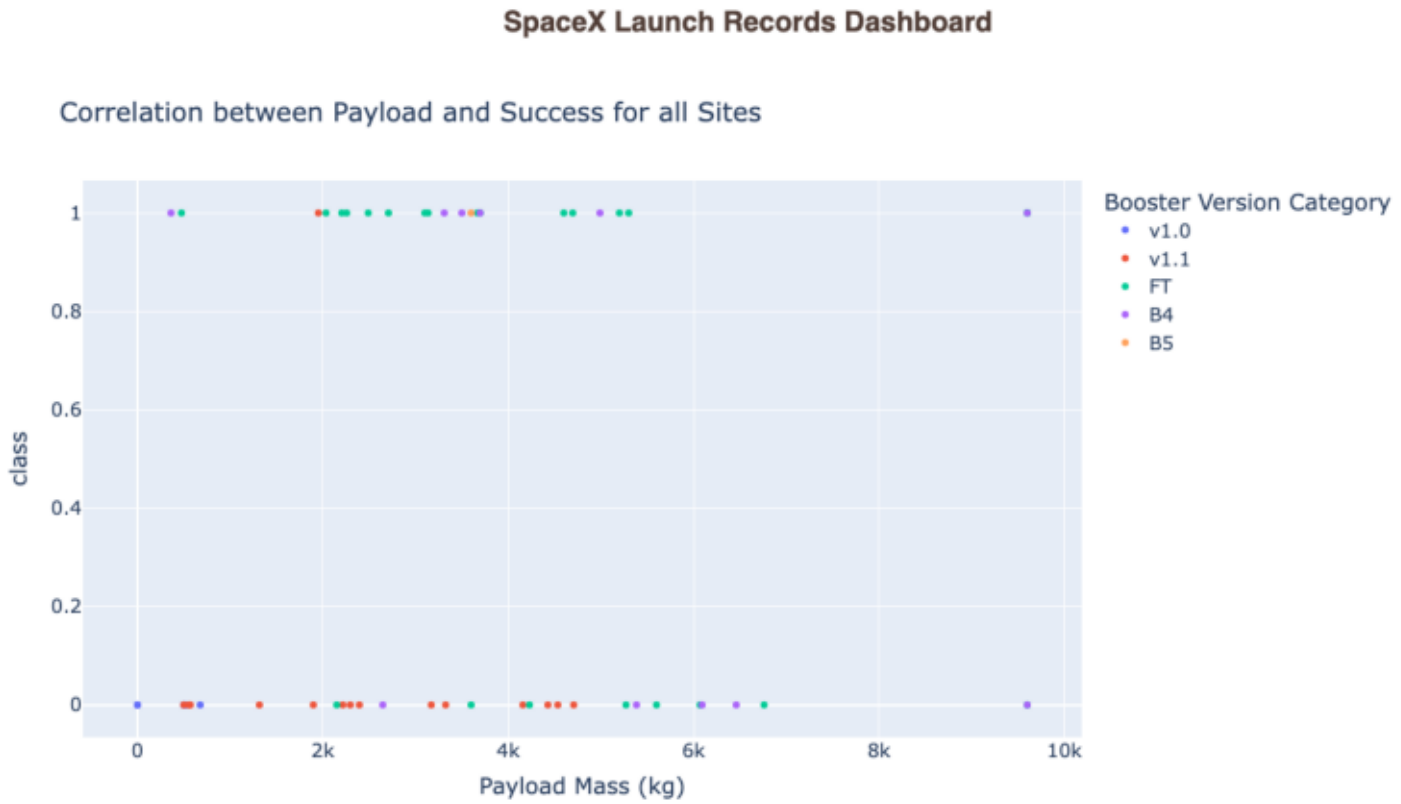
Site with Highest Launch Success Ratio – KSC LC-39A

DASH



- Dropdown = 'KSC LC-39A': success (1) vs failure (0).
- 76.9% success (10 of 13 launches) – the best ratio of all sites.
- This site hosts most of the newer FT/B4/B5 boosters.

Payload vs. Launch Outcome – All Sites

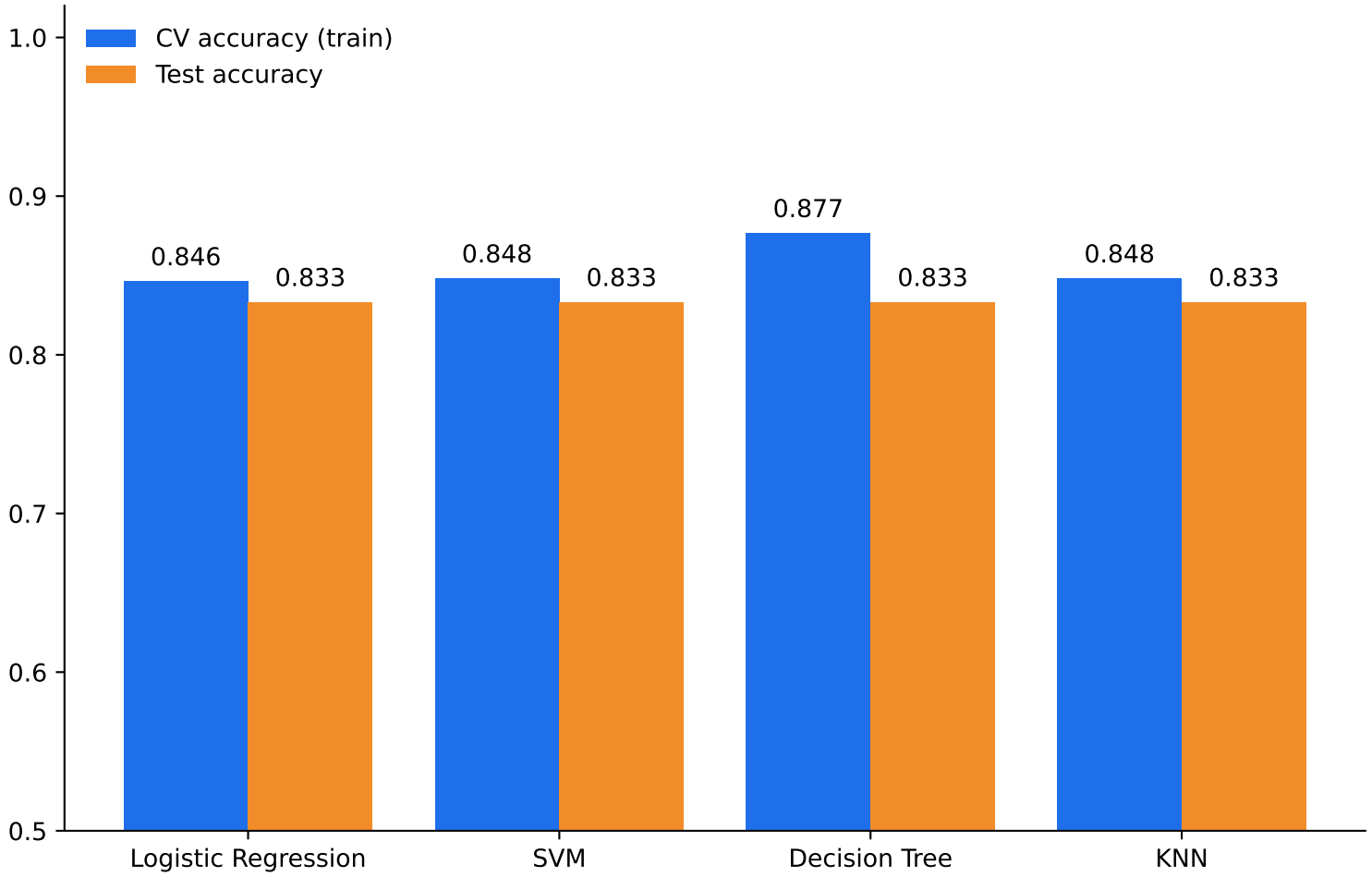


- RangeSlider 0–10,000 kg, points coloured by booster version.
- Payloads 2,000–5,500 kg have the highest success rate; > 5,500 kg few successes.
- FT booster has the most successes; v1.0 and v1.1 mostly failed.

Section 5

Predictive Analysis (Classification)

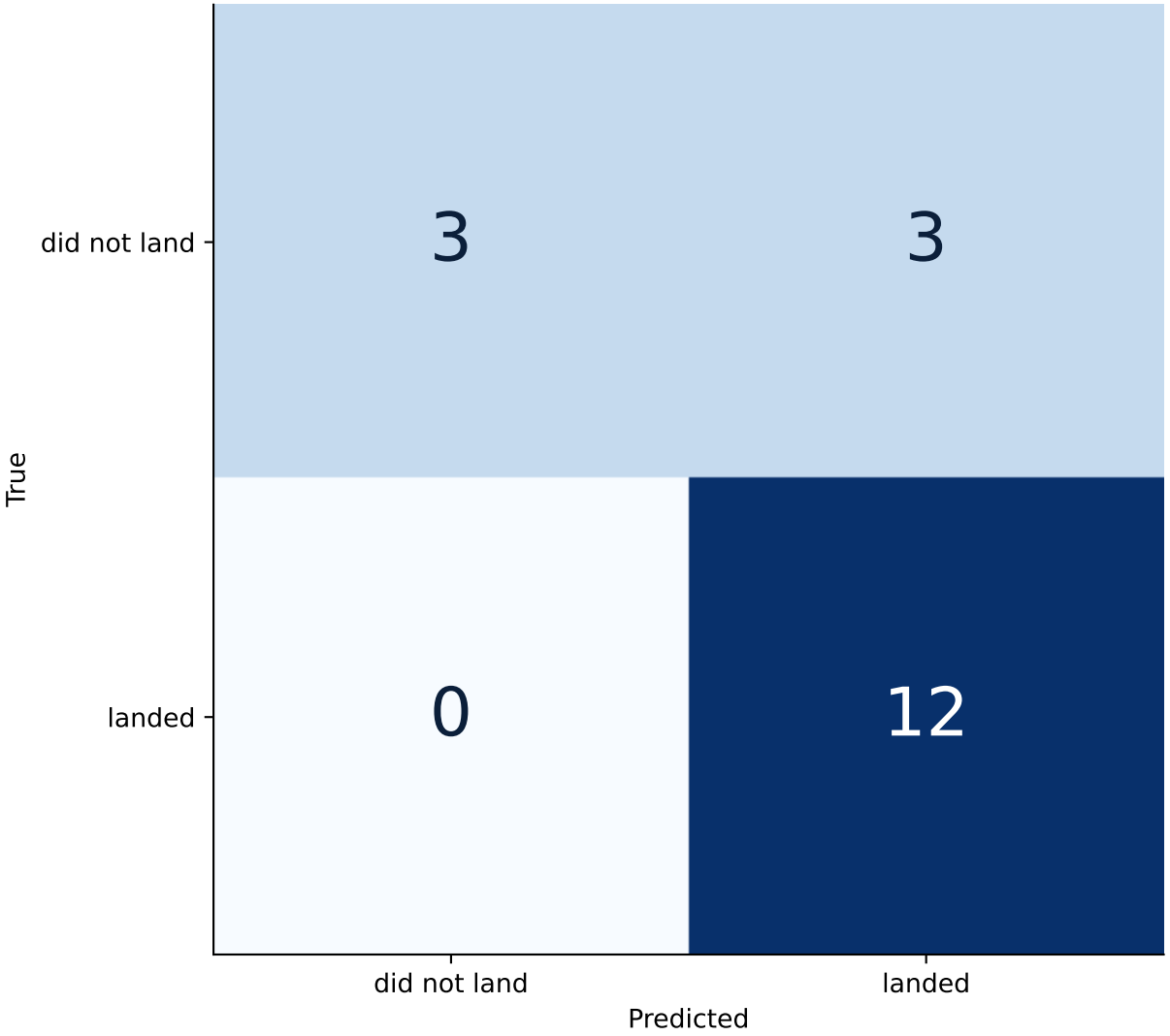
Classification Accuracy



Model	Best parameters
Logistic Regression	C=0.01, penalty=l2, solver=lbfgs
SVM	C=1, gamma=0.0316, kernel=sigmoid
Decision Tree	criterion=gini, max_depth=8, max_features=sqrt, min_samples_leaf=2, min_samples_split=10
KNN	algorithm=auto, n_neighbors=10, p=1

Insight: All models score 0.833 on the small test set (18 samples); Decision Tree has the best CV accuracy (0.877).

Confusion Matrix – Decision Tree



- True positives: 12, true negatives: 3
- False positives: 3, false negatives: 0
- The model distinguishes the classes well; the main problem is false positives — some failed landings are predicted as successful.
- Precision (landed): 0.80, Recall: 1.00

Conclusions

- The larger the flight number (experience), the higher the landing success rate at every launch site.
- Success rate rose from 0% in 2010–2013 to ~80%+ from 2017 onward.
- Orbits ES-L1, GEO, HEO and SSO have 100% success; GTO is the most difficult.
- KSC LC-39A has the most successful launches and the highest success ratio among sites.
- Launches with payloads below ~5 000 kg and FT / B4 / B5 boosters are more likely to land.
- Launch sites sit near the equator line and the coast, and far from cities.
- Decision Tree is the chosen predictor (83.3% test accuracy) — SpaceY can use it to estimate first-stage recovery and thus SpaceX's launch cost when bidding.

Innovative Insights

- Learning curve effect: success depends more on accumulated experience (flight number, block version, reuse count) than on payload alone — a competitor should expect SpaceX's cost advantage to keep growing.
- Test-set accuracy is identical for all models because the test set is only 18 launches; cross-validation accuracy is the more reliable criterion for model selection.
- Recommendation: retrain with newer launches (2021+) and add features such as booster reuse history and weather to reduce false positives.

Appendix

- GitHub repository (notebooks & Python files): https://github.com/TrungKiencding/IBM-Data-Science-Capstone-SpaceX
  - 1_jupyter-labs-spacex-data-collection-api.ipynb, 2_jupyter-labs-webscraping.ipynb, 3_labs-jupyter-spacex-Data-wrangling.ipynb
  - jupyter-labs-eda-sql-coursera_sqllite.ipynb, edadataviz.ipynb, lab_jupyter_launch_site_location.ipynb
  - spacex_dash_app.py, SpaceX_Machine_Learning_Prediction_Part_5.ipynb
- Datasets: dataset_part_1/2/3.csv, Spacex.csv, spacex_launch_geo.csv, spacex_launch_dash.csv (IBM Skills Network)
- Tools: Python, pandas, NumPy, Matplotlib, Seaborn, SQLite, Folium, Plotly Dash, scikit-learn

Thank you!